3D print request — Locus

From Shawki · hackathon demo prints · October 2026

Hi! Thanks a lot for helping with the prints. Here's what we need for our hackathon project.

What the project is: Locus makes maths graphs that blind students can read by touch. Every graph is a small flat plate with raised lines (axes and a curve) and a few braille dots for labels. We'll film hands touching the prints for our demo video.

What we need printed (all flat plates, no supports)

#	What	Size	File from us	Needed by
1	**Test piece**: pairs of raised lines at different gaps, a few line thicknesses and some braille dots, to check what your printer reproduces cleanly and what can be felt.	~100 × 40 mm	~16–17 Oct	~18–19 Oct
2	**Graph A**: the "bad" first version (lines too close together)	~100 × 100 mm	~21–22 Oct	~23 Oct
3	**Graph B**: the corrected final version	~100 × 100 mm	~21–22 Oct	~23 Oct
4–5	One spare each of Graph A and B, in case something breaks during filming	same	same	same

We'll send ready-made **STL files**. Please **print them at 100% scale and don't resize them.** The exact millimetres matter, because the whole point is that the gaps are wide enough to feel.

Suggested settings (you know your printer best, so change anything that won't work)

- **Material:** PLA
- **Layer height:** 0.12–0.16 mm (fine layers so the small braille dots come out round and crisp)
- **Base plate:** 2 mm thick, printed flat on the bed, no supports
- **Infill:** whatever you normally use. The plate is thin, so it'll be almost solid anyway.
- **Two colours if possible:** the plate one colour (e.g. white) and the raised lines and dots another (e.g. black). Use your slicer's "pause at height" / "filament change" right where the base ends, so the lines show up clearly on camera. If it's a hassle, one colour is fine.
- **Edges:** rounded or chamfered edges are nice, since people will be holding them, but not required.

The small details that matter most

- **Raised lines:** about 1.2 mm wide and 1 mm high
- **Braille dots:** about 1.5 mm across and 0.5–0.8 mm high, about 2.5 mm apart inside a letter
- **Smallest gap between two lines that must stay separate:** 3.2 mm

Could you tell us a few things before we make the files?

1. Printer model, nozzle size (0.4 mm?) and the largest flat area it can print
2. Which filament colours you have
3. The thinnest line and smallest dot your printer prints reliably
4. Roughly how long a ~100 × 100 mm thin plate takes to print
5. Whether it's OK to thank you by name in our project README and video credits (totally fine if not)

After the test piece, a quick photo and your honest opinion ("these dots came out blobby", "this gap merged") would help a lot. We'll use it to fix our settings before the final prints.

We'll also cover the filament cost. **Thanks so much!**